

A Multi-Tenant SaaS Platform for Community-Driven Hyper-Local Environmental Monitoring and Cleanup Coordination

Background

With the increasing impact of urbanization on local ecosystems, community-led environmental initiatives are becoming vital. In regions like Sri Lanka, local environmental protection such as for the Bolgoda Lake area, often relies on manual reporting and fragmented social media groups to organize cleanups or report pollution. Currently, there is a lack of centralized, user-friendly digital tools that allow citizens to report environmental issues (like illegal dumping or water pollution) and enable local NGOs or community groups to manage these "leads" as actionable tasks. Relying on generic messaging platforms leads to lost data, uncoordinated efforts, and a lack of long-term tracking for environmental restoration.

Problem Description

Community environmental groups face significant challenges in managing reports from the public. Reports of pollution or illegal activities are often buried in chat threads, making them difficult to track or verify. Existing environmental management systems are often designed for government agencies or large-scale enterprises, requiring complex GIS expertise and high costs. There is a clear need for a simple, mobile-first, and affordable SaaS platform that enables centralized reporting, task assignment for volunteers, and transparent progress tracking for small-scale environmental protection groups.

Proposed Solution

This project proposes the design and implementation of a Software-as-a-Service (SaaS) based environmental coordination platform. The system will centralize environmental "incidents" reported by the public into a single management interface for local administrators.

- Centralized reporting: citizens can submit reports with photos and geo-tags via a mobile app.
- Task management: administrators can convert these reports into cleanup "events" or "tasks" and assign them to registered volunteers.
- Multi-tenant architecture: the system will support multiple independent organizations (e.g., different local green societies) on the same platform.
- Configurable workflows: organizations can customize the status flow of a report (e.g., reported → verified → cleanup scheduled → resolved) without technical expertise.

Objectives

- To analyze the challenges local environmental groups face in coordinating volunteer efforts and tracking pollution reports.

- To design a centralized system for handling community-driven environmental incident reporting.
- To implement a full-stack platform (Mobile and Web) with real-time status updates and volunteer coordination features.
- To develop a multi-tenant backend that ensures data isolation for different environmental organizations.
- To evaluate the platform's effectiveness in reducing the time between incident reporting and community action.

Scope of the Project

The project includes an end-to-end system consisting of a mobile application for volunteers/citizens, a web application for administrators, and a robust backend.

Included in Scope: (can be refined as needed)

- Mobile application:
 - Primary interface for citizens to report incidents and volunteers to view assigned tasks.
 - Real-time notifications for new environmental issues in the user's vicinity.
 - Photo upload and map-based incident marking.
- Web application:
 - Administrative dashboard for organization leaders to manage reported incidents.
 - Volunteer database management and task assignment.
 - Basic reporting on cleanup progress and environmental impact metrics.
- Backend services:
 - Centralized database: storage for incidents, user roles, and organization-specific data.
 - Multi-tenant support: secure authentication ensuring users only see data belonging to their organization.
 - API layer: RESTful or GraphQL API supporting both mobile and web frontends.

Out of Scope:

- Automated satellite imagery analysis or Machine Learning for pollution detection.
- Integration with hardware IoT sensors for water or air quality monitoring.
- Financial transactions or donation processing systems.

Expected Outcomes

- A functional prototype of the Eco-Track platform.
- All the required non-functional properties should be made.
- A multi-tenant backend capable of hosting multiple environmental groups.
- Comprehensive technical documentation, including the database schema and system architecture.
- A project report analyzing how the platform improves coordination for a specific case study, such as a lake restoration project.